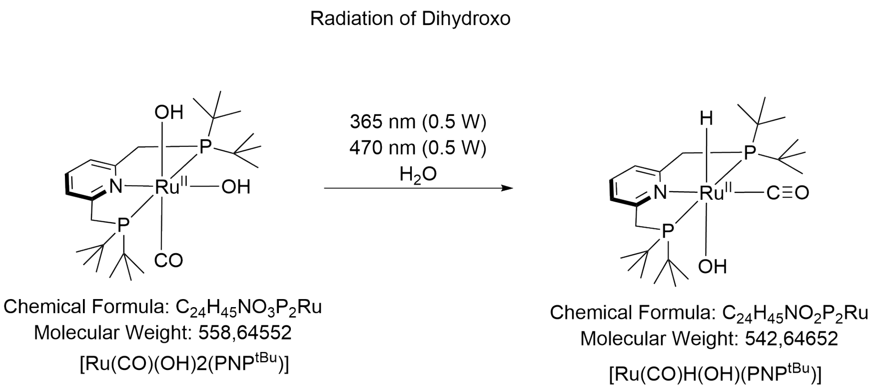
AE-617: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-610), kinetic

Date: 2025-11-11
Tags: Dihydroxo Radiation PNP P(tBu)N(py)P(tBu) [Ru(CO)(OH)2(PNP)] NMR AE 1H 31P Irradiation photocatalysis
Category: Two-photon
Status: Done
Created by: Alexander Eith

Objective

kinetic study of formation of hydride and side products during irrad of [Ru(CO)(OH)2(PNP{tBu})]

Reaction scheme/sample structure



ChemDraw File (linked): AE-590.cdxml

Literature/reference experiments

Literature	https://doi.org/10.1039/D1EE01053K
Reproduction	/
Similar experiments	Two-photon - AE-543: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-456-1), kinetic, max. conc Two-photon - AE-597: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-456-1), kinetic, max. conc

Reagents

Name	CAS Number / Experiment Number	Inventory number	Amount [mmol]	Equivalents	Mass$_{theo}$ [mg]	Mass$_{exp}$ [mg]	Molar mass [g/mol]	Density (g/ml)	Volume [ml]	Concentration [mM]	Obtained concetration [mM]
[Ru(CO)(OH)2(PNP{tBu})] (max. conc (approx. 1.44 mM) in water)	Two-photon - AE-610: Stocksolution of [Ru(CO)(OH)2(PNP{tBu})] in water	/	/	/	/	/	558.6455	/	0.60	max. conc (approx. 1.44 mM)	approx. 1.44

Irradiation Parameters

Power measurement was performed in experiment Two-photon - AE-616: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-610), kinetic

	Name
Used Set-up	Equipment - Advanced irradiation setup V1.0 I
Irradiation set-up number	Equipment - Irradiation setup 2 (CEEC II, E004)

	Light Source Name	Used power source	Wavelength [nm]	Power Setting [mW]	Analogue Setting [-]
First light source	Light Source - UHP LED 365 nm-2	Power Sources - BLS-18000-1 1	365	500	9.42
Second light source	Light Source - UHP LED 470 nm-2	Power Sources - BLS-13000-1E	470	500	6.97

Used beam combiner [Name or None]	Beam Combiner - Beam Combiner LCS-BC25-0409 -1
Irradiation distance [cm]	6.5
Thermostat temperature [°C]	25
Stirring speed [rpm]	/
Irradiation duration	1, 3, 20, 24 h

Procedure/observations

All steps (unless stated otherwise) were carried out under argon using standard Protocol - Schlenk Technique

The synthesis/reactions were performed in the dark by either wrapping the reaction flasks in aluminum foil or turning the lab light and light in the fume hood off, when the flasks were uncovered.

The NMR tube was placed in the back holder of the holder for 2 NMR tubes

Date	Time	Step	Observations	Pictures

06.11	14:55	A Young type NMR was prepared according to Protocol - Preparation of NMR Sample (from step 3) using 0.60 mL of **AE-610**, a Prep work - AE-489: Capillary production containg D2O and H3PO4	**AE-617**	/
	15:20	The NMR sample an NMR was measured	**AE-617-0H** slightly yellow solution	AE-617-0H.jpg
10.11	09:05	The sample was placed in the setup using the Photoreactor - AE-558: Design and print of holder and holder disk for 2 NMR tubes Holder_2NMR_tubes	no change	before irrad 1.jpg
	09:38	The irradiation was started	/	during irrad 1.jpg
	10:38	The irradiation was stopped after 1h	slligthly more orange	after irrad 1.jpg
	09:20	An NMR was measured	**AE-617-1H**	AE-617-1H.jpg
11.11	09:44	The sample was placed in the setup using the Photoreactor - AE-558: Design and print of holder and holder disk for 2 NMR tubes Holder_2NMR_tubes	no change	before irrad 2.jpg
	09:44	The irradiation was started	/	during irrad 2.jpg
	12:44	The irradiation was stopped after 3h	more orange	after irrad 2.jpg
	12:58	An NMR was measured	**AE-617-4H**	AE-617-4H.jpg
12.11	12:51	The sample was placed in the setup using the Photoreactor - AE-558: Design and print of holder and holder disk for 2 NMR tubes Holder_2NMR_tubes	no change	before irrad 3.jpg
	12:51	The irradiation was started	/	during irrad 3.jpg
13.11	08:51	The irradiation was stopped after 20h	no big change	after irrad 3.jpg
	09:01	An NMR was measured	**AE-617-24H**	AE-617-24H.jpg
	17:17	The sample was placed in the setup using the Photoreactor - AE-558: Design and print of holder and holder disk for 2 NMR tubes Holder_2NMR_tubes	no change	/
	17:18	The irradiation was started	/	/

14.11	17:21	The irradiation was stopped after 24h	no big change, maybe slightly darker	/
	17:39	An NMR was measured	**AE-617-48H**, no big change, maybe slightly darker	AE-617-48H.jpg

Analysis

Date	Time	Sample name	Analysis method	Analytical device	Solvent	Raw Data	Python script	Processed Data	Comparative Data	Interpretation
06.11	21:12	AE-617-0H	1H, 31P{1H}quant NMR	IAAC 400 MHz	H2O	AE-617-0H.zip	/	AE-617-0H_10.nmrium	/	As expected, before irrad, dihydroxo is main species
10.11	14:22	AE-617-1H	1H, 31P{1H}quant NMR	IAAC 400 MHz	H2O	AE-617-1H.zip	/	AE-617-0H_10.nmrium	/	No big difference
11.11	13:32	AE-617-4H	1H, 31P{1H}quant NMR	IAAC 400 MHz	H2O	AE-617-4H.zip	/	AE-617-0H_10.nmrium	/	Hydride observed
13.11	12:32	AE-617-24H	1H, 31P{1H}quant NMR	IAAC 400 MHz	H2O	AE-617-24H.zip	/	AE-617-0H_10.nmrium	/	Significant hydride formation
17.11	11:37	AE-617-48H	1H, 31P{1H}quant NMR	IAAC 400 MHz	H2O	AE-617-48H.zip	/	AE-617-48H_10.nmrium	/	Full conversion

Sample	Yield (hydride species, cis) [%]	Yield (hydride species, trans) [%]	Yield (dihydroxo) [%]	Yield (main side product, 57 ppm) [%]	Yield (soluable side products) [%]	Yield (other side products) [%]	Data
AE-617-1H	0	0	93	0	4	3	AE-617_Yield_Determination.xlsx
AE-617-4H	2	0	62	3	7	27	AE-617_Yield_Determination.xlsx
AE-617-24H	17	0	11	32	12	27	AE-617_Yield_Determination.xlsx
AE-617-48H	22	0	0	33	7	38	AE-617_Yield_Determination.xlsx

Results

Worked, significant product after 4 h, full conversion after 48 h, similar yields compared to Two-photon - AE-616: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-610), kinetic.

2 differences: higher conversion here after 4 h, otherwise conversion analog to AE-616

In AE-616 decrease in main side product from 24 to 48 h, here nearly constant

Linked experiments

Prep work - AE-489: Capillary production

Prep work - AE-549: Degassing of Milli-Q water

Two-photon - AE-597: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-570-1), kinetic, max. conc

Two-photon - AE-603: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-570-1), kinetic, max. conc, reproduction

Two-photon - AE-610: Stocksolution of [Ru(CO)(OH)2(PNP{tBu})] in water

Two-photon - AE-616: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-610), kinetic

Linked resources

Beam Combiner - Beam Combiner LCS-BC25-0409 -1

Equipment - Advanced irradiation setup V1.0 I

Equipment - Advanced power measurment setup V1.0 I

Equipment - Irradiation setup 2 (CEEC II, E004)

Light Source - UHP LED 365 nm-2

Light Source - UHP LED 470 nm-2

Power Meter - 843-R-USB + 919P-020-12

Power Sources - BLS-13000-1E

Power Sources - BLS-18000-1 1

Protocol - Schlenk Technique

Attached files

AE-617-48H.jpg
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AE-617_Yield_Determination.xlsx
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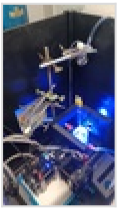

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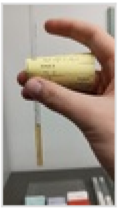

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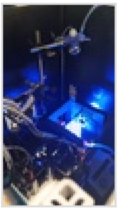

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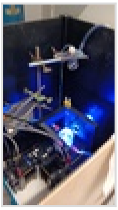

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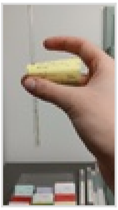

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